BMS Application Blueprint

System Navigation Architecture & Technical Specifications

Document Version: 1.0 **Project:** Business Management System (BMS) **Status:** Approved Architecture Spec

1. Purpose

This document defines the complete navigation architecture and screen relationships of the **Business Management System (BMS)**. It serves as the authoritative single source of truth for:

- Prototype implementation and interaction design
- UI navigation flows and user journeys
- Screen relationships and hierarchy
- Module boundaries and state transitions
- Future production development and routing implementation

All application navigation, workflows, and user interactions must strictly conform to this blueprint.

2. Application Structure

The BMS is organized into major functional modules structured hierarchically under a unified operational core:

```
Business Management System
│
├── Authentication & Onboarding
├── Dashboard
├── Point of Sale (POS)
├── Products
├── Inventory
├── Purchases
├── Suppliers
├── Customers
├── Sales
├── Reports
├── Users & Roles
├── Branch Management
├── Business Settings
└── System Support
```

3. Global Entry Points

The application defines four primary entry points based on user state and authentication context:

ENTRY POINT	USER TYPE	DESTINATION
Landing Page	Visitor	Login / Create Business
Invitation Link	Invited Employee	Invitation Flow
Login Page	Existing User	Dashboard
Password Reset Link	Existing User	Reset Password

4. Global Navigation

The flow chart below illustrates the primary global navigation pathways from entry to core application state:

```
Landing Page
│
├── Login
│       │
│       ▼
│   Dashboard
│
├── Create Business
│       │
│       ▼
│ Business Details
│       │
│       ▼
│ Workspace Selection
│       │
│       ▼
│ Dashboard
│
└── Forgot Password
        │
        ▼
Request Reset ──▶ Check Email ──▶ New Password ──▶ Success ──▶ Login
```

5. Invitation Flow

Invited users undergo a tailored onboarding process to ensure secure access to authorized organizational workspaces:

```
Invitation Link
      │
      ▼
Invitation Details
      │
      ▼
Security Setup
      │
      ▼
Invitation Accepted
      │
      ▼
Login ──▶ Dashboard
```

6. Dashboard

The **Dashboard** functions as the central navigation hub for all operational capabilities within the BMS:

```
Dashboard
│
├── POS            ├── Purchases        ├── Users
├── Products       ├── Suppliers        ├── Branches
├── Inventory      ├── Reports          ├── Settings
└── Customers      └────────────────────┴── Support
```

Each navigation menu item opens its corresponding functional module within the main content area while preserving context.

7. Module Architecture

To maintain UX consistency and predictable software patterns, every module follows a standard internal structure:

```
Module Root
│
├── List View      (Primary overview & filtering)
├── Details View   (Deep dive inspect mode)
├── Create Form    (Entity creation)
├── Edit Form      (Entity modification)
├── Delete Modal   (Confirmation & deletion)
└── History View   (Optional audit trail & change log)
```

Example Execution Pattern: Products Module

`Products List ▶ Product Details ▶ Edit Product ▶ Save ▶ Products List`

8. Navigation Rules

All UI components and route transitions must adhere strictly to these seven core navigation rules:

Rule 1: Clear Entry Points Every screen must have an explicit, unambiguous entry point.

Rule 2: Clear Exit Points Every screen must provide an obvious, visible exit mechanism (back, cancel, close).

Rule 3: No Dead Ends No screen or modal state may ever isolate or trap the user without a escape route.

Rule 4: Forward Primary Actions Primary actions (Save, Submit, Next) must advance the user forward in the flow.

Rule 5: Safe Secondary Actions Secondary actions (Cancel, Back) must return the user safely without data side-effects.

Rule 6: Graceful Error Recovery System errors and validation failures must highlight issues and never freeze or trap the user.

Rule 7: Feedback Guarantee Successful operations must always yield immediate visual confirmation (toasts, alerts, visual state changes).

9. Dashboard Navigation Rules

- **Persistent Shell:** The Dashboard shell and primary sidebar navigation are rendered once and never recreated during module transitions.
- **Persistent Navigation:** The sidebar remains visible and active at all times during authenticated sessions.
- **Workspace Swapping:** Module changes alter only the main content workspace region, maintaining UI state stability.

10. Authentication & Authorization Rules

System routing is strictly partitioned based on authentication state:

- **Public Routes (Unauthenticated):** Landing Page, Login, Forgot Password, Password Reset, Invitation Flow, Business Registration.
- **Protected Routes (Authenticated):** All other system views, modules, setting screens, and API endpoints require active session authentication.

11. Module Relationships

High-level conceptual interaction across business modules:

```
Authentication
        │
        ▼
Dashboard
        │
 ┌──────┼───────────────────────────────┐
 ▼      ▼           ▼         ▼         ▼
POS  Products   Inventory Customers Reports
 │      │           │         │         │
 └──────┴───────────┴─────────┴─────────┘
                  │
                  ▼
            Business Settings
```

12. Prototype Navigation Specifications

For the initial prototype implementation stage:

- Navigation is executed purely client-side (SPA state transitions).
- Authentication and session management are simulated via state flags.
- All operational data is mocked locally.
- Form inputs execute client-side validation logic only.
- No actual backend integration or external API calls are performed.

13. Future Production Notes

When transitioning from prototype to production deployment:

- **Authentication:** Simulated session flags will be upgraded to JWT (JSON Web Tokens) with secure storage and refresh mechanisms.
- **Data Hydration:** Mocked local data stores will be replaced with asynchronous RESTful / GraphQL API service calls.
- **Routing:** Client-side simulated state routing will transition to robust framework routing (React Router / Vue Router) with deep linking and history support.

- **Validation:** Client validation will be mirrored and strictly enforced by robust server-side validation layers.

14. Change Management & Architecture Rationale

This blueprint is a living architectural specification. All future screen additions, module extensions, or workflow modifications must be documented and approved in this blueprint before code implementation begins. No unmapped navigation paths may be introduced into the codebase.

> **Architectural Value Statement**
>
> This blueprint serves as the definitive architectural map of the BMS application. As each module from the Stitch export is reviewed, this document will be expanded with detailed screen-by-screen definitions, element destinations, dialog triggers, and edge-case handling. This disciplined approach guarantees predictable implementation, rapid QA testing, and effortless maintainability across the system lifecycle.